

# ***BookHeaven*** Software Requirements Specification (SRS)

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## **1. Introduction:**

### **1.1/ Purpose:**

- Purpose of this SRS document is giving developers a more clear picture of the concept

### **1.2/ Document Conventions:**

|     |                            |
|-----|----------------------------|
| NFR | Non-Functional Requirement |
| FR  | Functional Requirement     |
| US  | User Story                 |
| UC  | Use Case                   |

### **1.3/ Intended Audience:**

- Developers and Project Managers should read this document.

### **1.4/ Project Scope:**

#### **A. Will Do:**

- User registration and login/logout operations
- Determining the status of books (Read / Reading / To be Read)

#### **B. Won't Do:**

- E-Trade Feature(book buying/selling products)
- Automatic book recommendation system

## **2. Overall Description:**

### **2.1/ Product Perspective:**

- The system is planned to function independently and will not require integration with external systems. However, it can be extended in the future to interact with other platforms or APIs.
- Book Information APIs (e.g., Google Books API, Open Library API) could be used to automatically fetch book details such as author, cover image, and publication date.

### **2.2/ Product Functions:**

- The Book Tracker System provides the following major functions:
  - User registration and login/logout functionality
  - Profile creation and management
  - Secure authentication and password management
  - Add new books manually or (optionally) fetch details via an external API
  - Edit or delete existing books
  - Categorize books by status (Read / Reading / To Read)
  - Add personal notes, comments, or ratings for each book
  - View detailed book information (title, author, genre, publication year, etc.)

### **2.3/ User Classes and Characteristics:**

- There is only a single class of users, which is just users.
- Users will only need to know how to operate simple web applications.

### **2.4/ Operating Environment:**

- The BookHeaven Web Application is a browser-based system designed to run on a wide range of platforms and devices.

### **2.5/ Design and Implementation Details:**

- Personal Data Protection Law No. 6698 will be complied with and general ethical rules will be followed. There will be no other limitations or restrictions than this.

### **2.6/ Assumptions and Dependencies:**

- Assumptions:
  - Users have access to a stable internet connection and a chromium based web browser.

- Users possess basic computer literacy to navigate and interact with web applications.
- The system will be accessed primarily by individual users (not organizations or large-scale institutions).
- Dependencies:
  - Web Hosting Service for deploying the backend and frontend of the application.
  - Database Management System such as PostgreSQL for storing user and book data.
  - Web Browsers must comply with HTML5, CSS3, and JavaScript standards to ensure proper functionality.

## **3. Functional Requirements:**

### **3.1/ User Stories:**

#### **US1 / Adding Book Comments:**

- As a user
- I want to write a comment about one of my books
- So that i can express my opinion about that book.

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have added the book in question to my collection page.
- I have clicked on the book to access the comment menu.
- When I click submit, my comment should be added to the comments list.

#### **US2 / Adding Books to Collection Page:**

- As a user
- I want to add a book to my collection
- So that I can show the books I have an interest in.

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- When I click the "Add New Book" button a search bar and criteria selectors should appear.
- When I write keywords and press enter, top results should be shown.
- When I select criterias and press enter, top results should be shown.
- When I select the book I want to add, it should be added to my collection page.

### **US3/ Checking Comments:**

- As a user
- I want to get info about a book that i wanna read by using the comms of that book
- So that i can decide whether to read it or not

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have to search the name of the book that I wanna check by using the search bar.
- When i click to the comments, all of the comments will be shown.

### **US4/ Removing Books:**

- As a user
- I want to remove a book from my collection
- So that I can accurately display my collection.

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have to click to the "My Collection", then books in my collection will be shown.
- I have to right-click on the book that I want to remove.
- Then when I click the "Remove" button , the book will be removed.

### **US5/ Rating Books:**

- As a user
- I want to give a rating to one of my books
- So that I can express my opinions on that book.

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have to click to the "My Collection", then books in my collection will be shown.
- Then I have to click on the book that I want to rank.
- When I click the "Comments" and "Create a comment" buttons and write my opinions and click the "Submit" button. My comment will be submitted.

### **US6/ Checking Rankings Page:**

- As a user
- I want to check my rankings page
- So that I can see my books ranked from my least to most liked books.

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have to click to the "My Collection", then books in my collection will be shown.

### **US7/ Changing Profile Picture:**

- As a user
- I want to change my profile picture.
- So that I can express myself better.

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have to click on my Profile Picture to edit.
- When I click the "Edit profile picture" button, profile photos that I can make will be shown.
- Then I have to choose one of them to change my profile picture.

### **US8/ Changing Password:**

- As a user
- I want to change my password
- So that I can secure my account better.

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have to click the "Settings" button.
- Then I have to click the "Change Password" button
- I have to enter my password and a new password to change it.
- System has to send me a confirmation email if it was successful.

### **US9/ Logging In:**

- As a user
- I want to login
- So that I can access my profile.

#### **Acceptance Criteria:**

- I have to enter my username and Password in entrance page to log in

### **US10/ Toggling Sounds Off:**

- As a user
- I want to turn off the interface sound.
- So that i can browse my profile in silence

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have to click the "Settings" and "Sound" buttons.
- I have to click the "turn off" button which is placed next to "Interface Sound".

### **US11/ Reporting Comments:**

- As a user
- I want to report a user
- So that the site is kept civil.

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have to search for a book or select a book from my collection.
- I have to click the "Comments" button to see the comments.,
- When I see a comment containing hate speech etc. then i can report

### **US12/ Logging Out:**

- As a user
- I want to logout
- So that I can safely leave my profile.

#### **Acceptance Criteria:**

- I have to be logged in to my profile
- I have to click the "Log out" button.
- When I click the "Log out" button,I should be securely logged out of my profile.

### **US13/ Liking Comments:**

- As a user
- I want to drop like to the someone's comment
- So that I can show my support.

#### **Acceptance Criteria:**

- I have to be logged in to my profile.
- I have to have clicked a book on my collection page.
- I have to click the like button on a particular comment.
- When I click like, system should update the like amount on that comment.

### **US14/ Changing Book Status:**

- As a user
- I want to change my reading status on a particular book
- So that I can accurately showcase the books I have read.

#### **Acceptance Criteria:**

- I have to be logged in to my personal account.
- I have to click to the "My Collection", then books in my collection will be shown.
- I have to right-click to the book that i want to modify
- Then when I click the "Change Reading Status" button, a menu will open.
- When I select the correct status, system will change status accurately.

### **US15/ Updating “To-Read” List :**

- As a user
- I want to add a book I've been meaning to read to my to-read list
- So that i can memorize easily.

#### **Acceptance Criteria:**

- I have to be logged in to my profile
- I have to click “To-Read” button.
- When I click

### **3.2/ Use Cases:**

#### **UC1/ Adding & Removing Books:**

- Description: Users need to add and remove books from their collection page.
- Primary Actors: Users
- Stakeholders: Developers, Users
- Precondition:
  - User should be logged in and on their collection page.
- Main Flows:
  - Adding Flow:
    - User clicks the add book button present on the collection page.
    - System opens a book adding interface for the user.
    - User enters keywords and selects search criteria(e.g. Genre).
    - System returns top results for the data given by searching the database.
    - User selects the book that they are looking for.
    - System adds the book in question to the user's collection page.
  - Removing Flow:
    - User clicks the book on their collection page.
    - System opens book interface.
    - User clicks remove book button.
    - System removes the book from the user's collection page.

#### **UC2/ Register:**

- Description: Users need to create accounts for the website.
- Primary Actors: Users
- Stakeholders: Developers, Users

- **Precondition:**
  - User should be on the login page.
- **Main Flow:**
  - User clicks the register button on the login page.
  - System directs to a register menu.
  - User enters email and desired password.
  - If the information is valid and an account with the same email doesn't exist, a confirmation email is sent and user is directed to collection page.
  - If the information is invalid or an account with the same email exists, user is directed back to a blank register menu.

### **UC3/ Login & Logout:**

- **Description:** Users need to securely enter and exit their personal profiles.
- **Primary Actors:** Users
- **Stakeholders:** Developers, Users
- **Precondition:**
  - User should be on the log in page. (When doing Login operation)
  - User should click the logout button present on collection page. (When doing Logout operation)
- **Main Flows:**
  - **Login Flow:**
    - User enter their username and password and clicks login button.
    - If login information is valid, user is directed to their collection page.
    - If login information is invalid, give user feedback (e.g. "Username or Password is incorrect.").
  - **Logout Flow:**
    - User is asked for confirmation.
    - If confirmation is given, system securely logs out the user and directs user to login page.
    - If confirmation is cancelled, flow ends.

### **UC4/ Commenting:**

- **Description:** Users need to add comments to the comments list to the books on their collection.
- **Primary Actors:** Users
- **Stakeholders:** Developers, Users



- **Precondition:**
  - User should be logged in and on their collection page.
  - User has clicked one of the books on their collection page and opened the book interface.
- **Main Flows:**
  - User clicks on comment writing box present on the book interface.
  - User presses enter after writing desired text.
  - System adds comment to the comments list.
  - User's own comment is shown on top of comments list.

#### **UC5/ Ranking:**

- **Description:** Users need to add and remove books from their collection page.
- **Primary Actors:** Users
- **Stakeholders:** Developers, Users
- **Precondition:**
  - User should be logged in and on their collection page.
  - User has clicked one of the books on their collection page and opened the book interface.
- **Main Flows:**
  - User clicks the rate button.
  - User enters desired rating between 0 and 5, presses confirm button.
  - System updates book's rating in collection page.
  - System updates book's ranking present on the ranking page.

#### **UC6/ Reporting Comments :**

- **Description:** User wants to report an offending comment on a particular book's comments list.
- **Primary Actors:** Users
- **Stakeholders:** Developers, Users
- **Precondition:**
  - User should be logged in and on their collection page.
  - User should have clicked one of the books on their collection page and opened the book interface.
- **Main Flows:**
  - User clicks the report button present on each comment's box.
  - User selects the reporting reason on the menu that opens.
  - User clicks send and system sends a report to a report box.

### 3.3/ Functional Requirements List:

- **User Management Module:**
  - FR1.1: The system shall allow users to create a new account using a valid email address and password.
  - FR1.2: The system shall allow users to log in and log out securely.
  - FR1.3: The system shall allow users to reset their password if forgotten.
  - FR1.4: The system shall store user credentials securely using encryption or hashing.
  - FR1.5: The system shall allow users to update their profile information (e.g., name, avatar).
- **Book Management Module:**
  - FR2.1: The system shall allow users to add new books by entering details such as title, author, genre, and publication year.
  - FR2.2: The system shall allow users to delete books from their collection.
  - FR2.3: The system shall categorize books under three main statuses: "Read," "Reading," and "To Read."
- **Book Details and Notes Module:**
  - FR3.1: The system shall allow users to add personal notes or comments for each book.
  - FR3.2: The system shall allow users to rate books using a star scale.
  - FR3.3: The system shall display detailed information for each book (title, author, publication date, genre, user notes, rating).
  - FR3.4: The system shall allow users to edit or delete their notes and ratings.
- **Search and Filter Module:**
  - FR4.1: The system shall provide a search bar to find books by title, author, or keyword.
  - FR4.2: The system shall allow users to filter books by reading status, rating, or genre.
  - FR4.3: The system shall allow users to sort books alphabetically, by date added, or by rating.

- Reading Statistics Module:
  - FR5.1: The system shall track the number of books read by the user.
  - FR5.2: The system shall display reading progress and statistics (e.g., books read per month or year).
- System and Data Management Module:
  - FR6.1: The system shall store all data (users, books, notes) in a database.
  - FR6.2: The system shall validate input data before saving to the database.
  - FR6.3: The system shall ensure data persistence and prevent data loss.
  - FR6.4: The system shall support secure communication between client and server using HTTPS.
  - FR6.5: The system shall be responsive and accessible on desktop browsers.

## **4. Non-Functional Requirements:**

### **4.1/ Performance Requirements:**

- NFR1.1: The system shall load the main dashboard page within 2 seconds under normal network conditions.
- NFR1.2: Search and filtering functions shall return results within 2 seconds for up to 5,000 records.
- NFR1.3: The system shall support at least 100 concurrent users without noticeable performance degradation.
- NFR1.4: The database shall execute standard queries within 1 second on average.

### **4.2/ Security Requirements:**

- NFR2.1: All user credentials shall be stored securely using hashing or encryption.
- NFR2.2: The system shall use HTTPS for secure data transmission between client and server.
- NFR2.3: Only authenticated users shall be allowed to access their personal data.

### 4.3/ Usability Requirements:

- NFR3.1: The user interface shall be intuitive and easy to navigate for first-time users.
- NFR3.2: All primary features (add, search, update, delete) shall be accessible within 2–3 clicks.
- NFR3.3: The interface shall follow consistent color schemes, fonts, and layout standards for a pleasant user experience.

### 4.4/ Reliability and Availability Requirements:

- NFR4.1: The system shall maintain 90% uptime during regular operation.
- NFR4.2: The system shall preserve user data integrity during unexpected failures or restarts.

### 4.5/ Maintainability Requirements:

- NFR5.1: The codebase shall be modular and well-documented to simplify debugging and updates.
- NFR5.2: The system shall support version control (e.g., Git) for collaborative development.
- NFR5.3: The application shall allow easy integration of new features without major redesign.

### 4.6/ Compatibility Requirements:

- NFR6.1: The system shall be compatible with modern web browsers.
- NFR6.2: The application shall comply with HTML5, CSS3, and JavaScript ES6+ standards.

## **5. System Models:**

User Flow Diagram Link:

[https://github.com/memoryVoid1/BookHeaven\\_Deliverables/blob/main/D2\\_Images/SystemDiagramJpeg.jpg](https://github.com/memoryVoid1/BookHeaven_Deliverables/blob/main/D2_Images/SystemDiagramJpeg.jpg)

Use Case Diagram Link:

[https://github.com/memoryVoid1/BookHeaven\\_Deliverables/blob/main/D2\\_Images/UseCaseDiagram.jpg](https://github.com/memoryVoid1/BookHeaven_Deliverables/blob/main/D2_Images/UseCaseDiagram.jpg)